Ziphozinle Mqopi

40
40
100%

ELECTRICITY AND ELECTRONICS

FORMATIVE ASSESSMENT

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Ziphozihle Maqpi	Manager:
Trainer:	Mentor / Coach:
Facilitator: Balerinah Yabadi	
Assessor: Balerinah Yabadi	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozihle Maqpi

Learner's signature: _____ Date: _____

Facilitator Name: Balerinah Yabadi

Facilitator's signature: _____ Date: _____

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

_____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

EASTC TECHNOCENTRIC VARSITY

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozinhle Maqopi	Signature
Facilitator Name: Balenaah Tjabadi	Signature:

EASTC TECHNOCENTRIC VARSITY

LEARNING UNIT 1:
KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

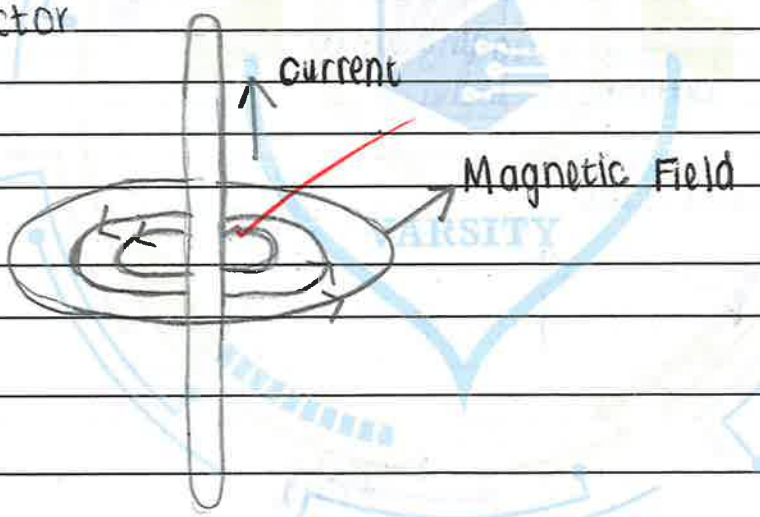
Task	Questions Description	Marks
1.	a. Describe, calculate and interpret fundamental concepts of electricity (Electro motive force, Potential difference, Resistance)using the correct units of measurement and definitions. b. Describe the factors that influences the resistance of a material c. Define and explain, using the correct units of measurement ohms law of electricity d. Define and explain using the correct units of measurement Kirchoff's law of electricity Manipulate formula to calculate voltage, current and resistance in series/parallel circuits	**10**

10) **EMF** : The eletromotive force is the energy provided by a cell or battery to ~~drive~~ drive electric current through a circuit. It is the pontential difference between the terminals of a cell or battery when no current is flowing. Measured in volts (V)

Task	Questions Description	Marks
2	a) Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field. Fleming's left-hand rule must be demonstrated. b) Describe and explain the various forces or methods that can be used to alter magnetic fields, and describe the changes that take place	10

2a) When an electric current flows through a conductor, it generates a magnetic field around it

Magnetic field around around a current-carrying conductor



Interaction with a Uniform Magnetic Field

When the current-carrying conductor is placed in a uniform magnetic field, the two magnetic field interact. The magnetic field around the conductor interacts with the external magnetic field, resulting in a force acting on the conductor

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LEARNING UNIT 2:
KM-03-KT03: Earthing and bonding (35%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
3	a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation b) Explain the term 'systems earthing' c) Describe the earthing systems relevant to SANS 10142-1. d) Describe how a common earth electrode is used in reticulation circuits	10

3a) The regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation are crucial for ensuring safety and compliance. Supplier side: The supplier must establish a reliable earthing system that minimizes the risk of electric shock and equipment failures. Consumer side: The consumer must connect neutrals to earth at service panels in accordance with local codes. This involves ensuring that the neutral conductor is properly connected to the earth at the main service panel to ensure that in case of a fault, current will safely return to the ground

b) System earthing, also known as neutral earthing, involves

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Fault currents, thereby protecting people and equipment from electrical shocks and damage. The Earth Electrode is typically connected to the neutral point of the distribution transformer, effectively bonding the electrical system to the Earth. This connection is further extended to the metal parts of equipments and structures, creating a comprehensive network that ensures all components at the same electrical potential as the earth. In the event of a fault, such as a short circuit to a metal enclosure, the Earth electrode provides a low-impendence path for the fault current to ~~fall~~ flow safely to ground, triggering protective devices like circuit breakers or fuses to isolate the fault



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capacitance

capacitance values. Application: Widely used in power supply circuits for smoothing and filtering, coupling and decoupling applications, and in audio equipment to block DC while allowing AC signals to pass

b) Calculating capacitance in series and parallel

Capacitors in Series

$$\frac{1}{C_{total}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \cdots$$

if you have two capacitors $C_1 = 4\,\mu F$ and $C_2 = 6\,\mu F$ the total capacitance is calculated as follows

$$\frac{1}{C_{total}} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12} = \frac{12}{5} = 2,4\,\mu F$$

10

Capacitors in Parallel

$C_{total} = C_1 + C_2 + C_3 + \cdots$

For two capacitors $C_1 = 4\,\mu F$ and $C_2 = 6\,\mu F$

$C_{total} = 4\,\mu F + 6\,\mu F = 10\,\mu$

c)

Colour	Digit	Multiplier
Black	0	X1
Brown	1	X10
Red	2	X100
Orange	3	X1k
Yellow	4	X10k
Green	5	X100k
Blue	6	X 1m
Violet	7	X 10M
Grey	8	
White	9	
Gold		±5%
Silver		±10%

Resistor Colour Code Calculation:

- Brown = 1
- Black = 0
- Red = 10^2 (100)
- Gold = ±5% tolerance

Value: 10 x 100 = 1kΩ ±5%

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ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS
Good work

ASSESSMENT JUDGEMENT
Learner's Total Mark: 100 **Requirements met** ☒ **Requirements not met** ☐
Action/s required: **By when:**

LEARNER FEEDBACK & COMMENTS
~~Exellent~~ Excellent

DECLARATION BY THE FACILITATOR
I Baienaah Tjabadi (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

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Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Assessment Planning

	Theory	Practical
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments

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